

Introduction

The problem being solved with big data analytics in weather forecasting is the need for more accurate and timely weather predictions. With the increasing availability of data from various sources such as weather stations, satellites, and sensors, there is a need for advanced analytics techniques to process and analyze this data to generate accurate weather forecasts.

The problem is being solved by using big data analytics techniques such as data mining, machine learning, and artificial intelligence to process and analyze large volumes of weather data. These techniques can help identify patterns and trends in the data that can be used to generate more accurate weather forecasts.

The targeted domain for big data analytics in weather forecasting is the meteorological industry, which includes weather forecasting organizations, government agencies, and private companies that provide weather-related services. The use of big data analytics in weather forecasting can help these organizations and companies provide more accurate and timely weather predictions, which can be used for a variety of applications such as disaster management, agriculture, aviation, and transportation.

In terms of experimental setup, the process typically involves collecting and integrating data from various sources, cleaning and preprocessing the data, and then applying big data analytics techniques to analyze the data and generate weather forecasts. Suitable data visualization techniques such as graphs, charts, and maps can be used to present the findings in an easy-to-understand format.

The discussion and analysis of the findings would involve interpreting the results of the big data analytics techniques and evaluating their effectiveness in generating accurate weather forecasts. This can involve comparing the results with existing weather forecasts and assessing the impact of the new forecasts on decision-making processes in various applications.

Background of the study

Generic Information: Big data analytics is a process of examining large and complex datasets to uncover patterns and correlations, which can help organizations make informed decisions.

Problem Statement: The complexity and unpredictability of weather patterns pose significant challenges for accurate weather forecasting. Conventional methods often struggle to process vast amounts of weather data, leading to less precise predictions.

Aim/objective of the Report: This report aims to explore the application of big data analytics in weather forecasting, focusing on improving the accuracy and localization of weather predictions.

Contributions of the work connected with Methodology: The study employs advanced analytics techniques, including predictive models and statistical algorithms, to process diverse weather datasets. The methodology includes real-time data processing, machine learning algorithms, and data visualization techniques to provide actionable insights for weather forecasting.

Social/Research community Impact of the work: By leveraging big data analytics, the study contributes to the development of more accurate and localized weather forecasting systems. This can significantly benefit various sectors, such as agriculture, transportation, and emergency management, by enabling better decision-making and disaster preparedness. The findings can also foster further research in the field of big data analytics and weather forecasting, leading to advancements in data-driven decision-making processes.

Related Work

In the field of big data analytics in weather forecasting, there are several existing works that have made significant contributions. These works primarily focus on improving the accuracy and localization of weather predictions by processing vast amounts of weather data using advanced analytics techniques, such as predictive models and statistical algorithms.

However, our work differs from existing ones in several ways. Firstly, we employ a unique methodology that combines real-time data processing, machine learning algorithms, and data visualization techniques to provide actionable insights for weather forecasting. Secondly, our study focuses on a more comprehensive set of weather variables, including temperature, precipitation, wind speed, and humidity, to offer more accurate and localized weather predictions. Lastly, we aim to evaluate the social and research community impact of our work by conducting surveys and interviews with stakeholders in various sectors, such as agriculture, transportation, and emergency management, to understand how our findings can be applied to improve decision-making and disaster preparedness.

Methodology

Data Collection: The first step in the process is to collect data from various sources such as weather stations, satellites, radar, aircraft, and ships. This data can be structured or unstructured and can include information such as temperature, humidity, wind speed, and precipitation.

Data Cleaning and Preprocessing: Once the data is collected, it needs to be cleaned and preprocessed to remove any errors, inconsistencies, or missing values. This step is crucial to ensure the accuracy of the weather forecasts.

Data Integration: After cleaning and preprocessing, the data is integrated into a single database for analysis. This step involves resolving any issues related to data formatting, naming conventions, and units of measurement.

Data Analysis: The integrated data is then analyzed using various statistical and machine learning techniques to identify patterns and trends. This step involves using big data analytics tools and technologies such as Hadoop, Spark, and Storm.

Model Development: Based on the analysis, a weather forecasting model is developed. This model can be a physical model based on the laws of physics or a statistical model based on historical data.

Model Validation and Testing: The developed model is then validated and tested using historical and real-time data. This step is crucial to ensure the accuracy and reliability of the weather forecasts.

Deployment and Monitoring: Once the model is validated and tested, it is deployed for real-time weather forecasting. The model is continuously monitored and updated to ensure its accuracy and reliability.

In terms of data cleaning, it involves removing any errors, inconsistencies, or missing values from the collected data. This step is crucial to ensure the accuracy of the weather forecasts.

As for the chosen model, the choice of model depends on various factors such as the type of data, the accuracy requirements, and the computational resources available. Some commonly used models in weather forecasting include the numerical weather prediction (NWP) model, the statistical postprocessing (SPP) model, and the machine learning (ML) model.

Result and Discussion

Experimental Setup: The experimental setup for big data analytics in weather forecasting typically involves collecting large amounts of weather data from various sources, such as satellites, weather stations, and radar. This data is then cleaned and preprocessed to remove any errors or inconsistencies. Once the data is cleaned, it is integrated into a single database for analysis.

Discussion and analysis of the findings: The analysis of the weather data using big data analytics techniques can reveal hidden patterns and trends that can be used to improve weather forecasting. For example, machine learning algorithms can be used to analyze historical weather data and make predictions about future weather patterns. These predictions can then be visualized using data visualization techniques such as graphs and charts to make them easier to understand.

One study published in the Archives of Computational Methods in Engineering used a Systematic Literature Review (SLR) method to analyze big data analytic approaches in weather forecasting. The study found that big data analytics can help to solve challenges related to traditional data management techniques and technology in weather forecasting. The study also proposed a feasible taxonomy of the current reviewed papers as technique-based, technology-based, and hybrid approaches.

Another study published in the International Journal of Scientific Research in Science and Technology used Apache Pig, Hadoop Distributed File System, and Apache Hive to process large amounts of weather data obtained from the National Climatic Data Center. The study was able to explain the dataset by using given methods and outputting the maximum, minimum, and average temperature according to the given time and date.

A paper published in the IJSRD - International Journal for Scientific Research and Development reviewed big data analytics for weather prediction and studied the benefits of using it. The paper found that big data analytics can help to predict weather patterns more accurately and efficiently.

In the proposed system, a Naïve Bayse algorithm was used to predict earthquakes, floods, and other weather-related events. The system focuses on meteorological data to predict the seasons and separate the weather data based on longitudinal and latitudinal coordinates. This can be used to analyze the reliability factor of cyclones, earthquakes, rainfall, temperature, and humidity.

In a paper published in TEST Engineering & Management, a Pre-Processing Framework was proposed to address quality of data in weather monitoring and forecasting applications. The framework considers global warming parameters and raises alerts/notifications to warn users and scientists in advance.

In a paper published in IJCSMC, a Big Data prediction framework for weather temperature was built based on the MapReduce algorithm. The framework uses Information Mining Package to perform managed grouping methodology on immense measures of information.

In the IOCAG 2022 paper, a Big Data prediction framework for weather temperature was built based on the MapReduce algorithm. The framework uses Support Vector Machines (SVM) to perform classification and clustering of weather data.

In conclusion, big data analytics can help to improve weather forecasting by revealing hidden patterns and trends in large amounts of weather data. Various studies have shown that big data analytics can help to solve challenges related to traditional data management techniques and technology in weather forecasting. Data visualization techniques can be used to make the predictions easier to understand.